

Lesson 2 · Practice sheet

NCEA Level 2 · AS 91262 · Turning points and optimisation

Time: 30 minutes

Name: _____

Date: _____

Show all working. Every maximum or minimum needs a reason; every answer in context needs units. [A]
Achieved, [M] Merit, [E] Excellence.

1. Find the turning points of $y = x^3 - 27x + 10$ and justify their nature. [A]

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2. A stone is thrown upwards. Its height is $h = 1.5 + 14.7t - 4.9t^2$ metres after t seconds. Find its maximum height. [A]

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Question 3 [A]

- (c) Use calculus to find the values of x for which the graph of the function $f(x) = \frac{2x^3}{3} + \frac{3x^2}{2} - 20x - 3$ is decreasing.

Question 4 [A] · Merit with the justification

- (b) A cricket match is being played. A cricketer hits the ball, and its height, h , above the grass cricket pitch in metres, can be modelled by the function:

$$h = 22.5t - 4.9t^2 + 1$$

where t is the time in seconds since the ball was hit.

Using calculus methods, find the maximum height above the ground that the ball reaches.

Question 5 (i) [A] · (ii) [M]

- (c) The number of daily viewers for a new presenter on a video gaming streaming service can be modelled by the following equation:

$$V = -11t^2 + 528t \quad \{0 \leq t \leq 48\}$$

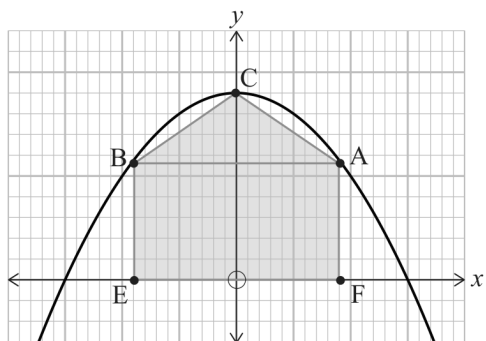
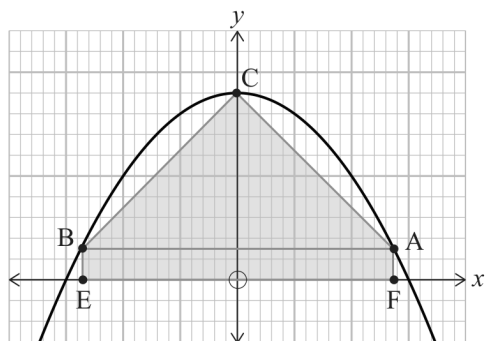
Where V represents the number of daily viewers and t represents time in months.

- (i) Find the rate(s) of change of daily viewers, with respect to t , when the number of daily viewers is 3520.

- (ii) What is the maximum number of daily viewers that this new presenter gets during the first 48 months?

(d) Two possible configurations of a house shape drawn below a parabola are shown below.

- C is at (0,9).
- A and B are points on the parabola $y = 9 - x^2$.
- E and F are on the x -axis.



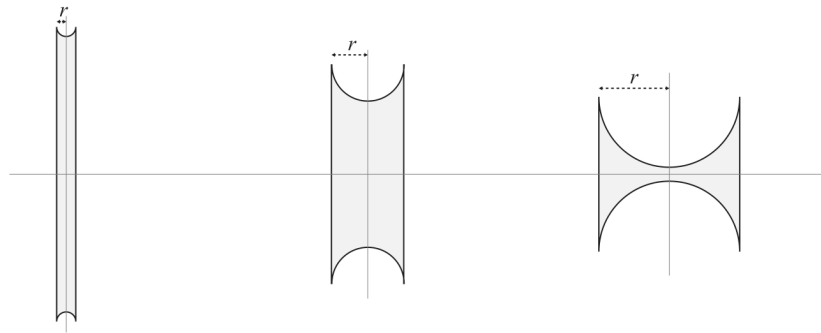
Question 7 Extension [E]

- (e) A new business is designing a logo based on a stylised 'H' shape, as seen in the diagrams.

The logo is to be designed with two semi-circles and two straight lines. The owner of the new business wants to include space for the company's name inside the shape, so he has asked the designer to maximise the area inside the shape.

The owner plans to build a replica of the logo, and due to material constraints, the total perimeter of the shape must not be greater than 80 cm.

Three examples are shown here:



- (i) Find the maximum area.
- (ii) Use calculus methods to show that this is a maximum.

Circumference of a circle: $C = 2\pi r$

Area of a circle : $A = \pi r^2$